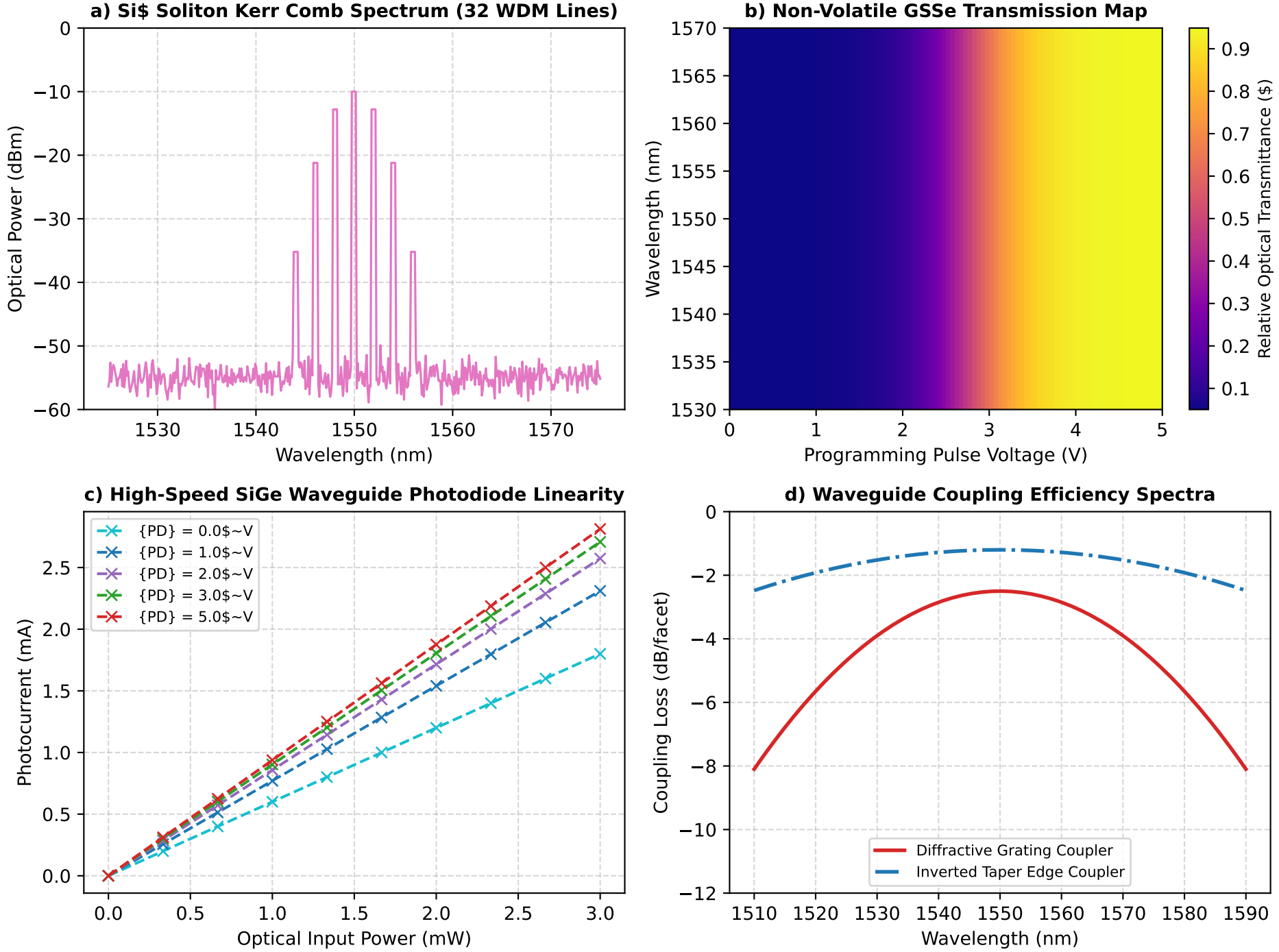

a) Si$ Soliton Kerr Comb Spectrum (32 WDM Lines)
Optical Power (dBm)
Wavelength (nm)
b) Non-Volatile GSSe Transmission Map
Wavelength (nm)
Programming Pulse Voltage (V)
Relative Optical Transmittance ($)
c) High-Speed SiGe Waveguide Photodiode Linearity
{PD} = 0.0$~V
{PD} = 1.0$~V
{PD} = 2.0$~V
{PD} = 3.0$~V
{PD} = 5.0$~V
Photocurrent (mA)
Optical Input Power (mW)
d) Waveguide Coupling Efficiency Spectra
Coupling Loss (dB/facet)
Diffractive Grating Coupler
Inverted Taper Edge Coupler
Wavelength (nm)